

Review

Physics-Informed Machine Learning for Intelligent Gas Turbine Digital Twins: A Review

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Abstract

This review surveys recent progress in hybrid artificial intelligence (AI) approaches for gas turbine intelligent digital twins, with an emphasis on models that integrate physics-based simulations and machine learning. The main contribution is the introduction of a structured classification of hybrid AI methods tailored to gas turbine applications, the development of a novel comparative maturity framework, and the proposal of a layered roadmap for integration. The classification organizes hybrid AI approaches into four categories: (1) artificial neural network (ANN)-augmented thermodynamic models, (2) physics-integrated operational architectures, (3) physics-constrained neural networks (PcNNs) with computational fluid dynamics (CFD) surrogates, and (4) generative and model discovery approaches. The maturity framework evaluates these categories across five criteria: data dependency, interpretability, deployment complexity, workflow integration, and real-time capability. Industrial case studies—including General Electric (GE) Vernova’s SmartSignal, Siemens’ Autonomous Turbine Operation and Maintenance (ATOM), and the Electric Power Research Institute (EPRI) turbine digital twin—illustrate applications in real-time diagnostics, predictive maintenance, and performance optimization. Together, the classification and maturity framework provide the means for systematic assessment of hybrid AI methods in gas turbine intelligent digital twins. The review concludes by identifying key challenges and outlining a roadmap for the future development of scalable, interpretable, and operationally robust intelligent digital twins for gas turbines.

Keywords: physics-informed machine learning; hybrid modeling; gas turbine diagnostics; artificial neural networks; intelligent digital twin; virtual sensing; remaining useful life; transfer learning; generative models



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1. Introduction

Gas turbines remain a cornerstone of modern energy infrastructure, delivering both primary and dispatchable power generation while also supporting propulsion systems and industrial mechanical drives. As energy systems transition toward decarbonization and incorporate a larger share of renewable sources, the role of gas turbines is evolving. First, they are increasingly being adapted to operate on carbon-free fuels such as hydrogen and ammonia, as well as carbon-neutral synthetic fuels and low-carbon biofuels. Second, they are playing a growing role as reliable backup power sources for intermittent renewables [1,2]. The shift from steady baseload operation to flexible service, characterized

by rapid load changes, frequent cycling, and extended part-load operation, has introduced new operational challenges, particularly for aging fleets. Under these conditions, multiple degradation mechanisms act simultaneously, at varying rates, including (but not limited to) fatigue crack growth, hot corrosion, erosion, tip clearance increases, and blade twist. Together, these effects reduce aerodynamic performance and overall efficiency. Frequent load fluctuations also increase the occurrence of thermal–mechanical stress cycles, making traditional degradation patterns less predictable and accelerating the overall deterioration process. These evolving operational demands highlight the need for advanced monitoring and predictive capabilities. Hybrid AI-based digital twins, which combine physics-based degradation models with operational data, can enable proactive maintenance, optimize performance, and extend the service life of critical turbomachinery assets [2–5].

The concept of the digital twin, detailed in [6], describes a virtual representation of a physical asset, system, or process that mirrors its state and behavior in real time, enabling monitoring, simulation, and analysis throughout its lifecycle. This definition has since evolved into structured frameworks for product and system design that integrate multi-source data and lifecycle considerations [7,8]. Building on this foundation, the intelligent digital twin integrates artificial intelligence, machine learning, and advanced analytics to enable learning from multi-source data, adaptive behavior under changing operating conditions, and support predictive decision-making [9]. This progression from descriptive and predictive modeling toward adaptive, autonomous operation represents a critical step in applying digital twin technology for complex, high-performance systems such as gas turbines [10,11].

Prior reviews have provided insights into emerging physics-informed methods and data-driven tools but remain limited in scope and or in application. For instance, prior studies have focused primarily on the mathematical aspects of physics-informed neural networks (PINNs) applied to fluid mechanics, or surveyed machine learning methods in turbomachinery without emphasizing digital twin implementations or hybrid architectures [12–14]. Other surveys examined physics-informed ML from a methodological perspective but did not provide maturity assessments or detailed industrial case studies [12–14]. This review distinguishes itself by integrating industrial with four categories of hybrid AI methods and a comparative maturity framework tailored to gas turbine applications. This combination of academic rigor and industrial context defines the novelty of this work.

Conventional physics-based diagnostic methods, such as thermodynamic cycle simulations, component maps, and gas path analysis (GPA) [12–14], are interpretable and reproducible but computationally intensive and often limited in adapting to degraded and/or transient conditions. Purely data-driven approaches, particularly those using artificial neural networks (ANNs), capture nonlinear relationships and enable fast inference but struggle to generalize beyond training domains and lack physical interpretability [14–17]. These limitations have motivated the emergence of hybrid artificial intelligence (AI) or physics-informed machine learning (PIML) methods, which embed physical constraints, simulation outputs, or governing equations into learning architectures. Hybrid frameworks, such as ANN–GPA combinations, have demonstrated improved sensitivity and robustness in degradation tracking [14,15], though they remain challenged by sparse data, limited fault representation, and the need to maintain alignment with physical laws [16–23].

Hybrid AI research for gas turbine modeling is progressing along several trajectories that share common mathematical and computational foundations. Equation discovery methods, such as Sparse Identification of Nonlinear Dynamics (SINDy) and its open-source implementation PySINDy, extract governing equations directly from data [24]. Physics-constrained networks, including physics-informed neural networks and Navier–Stokes

Flow Net (NSFnet) variants, embed conservation laws into loss functions for laminar and turbulent flows, with residual-based attention improving convergence [5,25–27]. Operator learning frameworks, such as the Fourier Neural Operator (FNO) and Fourier DeepONet, learn resolution-invariant mappings for partial differential equations (PDEs), enabling scalable, mesh-independent aerothermal predictions [28–32]. Architectures preserving physical invariants, such as Lagrangian Neural Networks (LNNs) and Graph Neural Networks (GNNs), extend inference to coupled thermofluid–structural systems [33–38].

Physics-informed machine learning methods have also been applied to high-fidelity thermal processes relevant to gas turbine operation and manufacturing. In coating and additive manufacturing, they predict transient temperature fields critical for microstructural integrity and the remaining useful life (RUL) of hot-section components [39]. In forced convection systems, physics-informed architectures achieve real-time thermal predictions by embedding governing heat-transfer equations into neural networks, enabling fast, physically consistent simulations under changing flow conditions [40]. These capabilities are directly applicable to intelligent digital twins and their embedded sub-simulations, enhancing preventive and predictive maintenance strategies and component life assessments.

Building on these distinctions, Section 2 introduces four categories of hybrid AI methods, Section 3 presents the comparative maturity framework, Section 4 discusses industrial relevance and future trends, Section 5 outlines the limitations of this review, and Section 6 concludes with recommendations.

2. Hybrid AI Methodologies for Gas Turbine Applications: Description, Advantages and Limitations

Hybrid artificial intelligence (AI) for gas turbine diagnostics, predictive maintenance, and performance optimization combines the strengths of data-driven learning and physics-based modeling, overcoming the limitations each approach faces when applied independently. Earlier reviews have organized hybrid AI primarily by network type—for example, Physics-Informed Neural Networks (PINNs) and operator-learning frameworks [17,28,31,32]—by application area such as diagnostics and monitoring [19,39,40], or by the general level of physics integration [4,41].

In this review, categories are instead defined by the specific stage in the modeling architecture where physics and AI interact. This perspective clarifies whether physics enters at the data level (e.g., physics-augmented training datasets), within the learning architecture itself (e.g., PINNs), or at the system and decision layer (e.g., optimization frameworks and fleet-level digital twins). Based on this integration logic, four distinct categories of hybrid methods are identified:

- ANN-augmented thermodynamic models;
- Physics-integrated operational architectures;
- Physics-constrained neural networks and Computational Fluid Dynamics surrogates;
- Generative and model discovery approaches.

Each category presents distinct trade-offs in terms of generalizability, accuracy, computational cost, and suitability for real-time implementation. This provides a comprehensive baseline for leveraging different methods in the development of intelligent digital twins. Earlier reviews have been more limited in scope, often focusing on specific aspects. For example, some organized hybrid AI by network type, such as Physics-Informed Neural Networks (PINNs) [17,28,31,32] or operator-learning frameworks [19,39,40]; others classified approaches by application area (e.g., diagnostics, monitoring) or by the general level of physics integration [4,41].

2.1. ANN-Augmented Thermodynamic Models

ANN-augmented thermodynamic models couple classical thermodynamic simulations and gas path analysis with artificial neural networks (ANNs) to improve diagnostic sensitivity, robustness, and real-time applicability. These methods are widely used for component-level diagnostics, degradation tracking, and health assessment [41]. ANN modules learn nonlinear relationships from synthetic or historical datasets, enabling rapid inference while retaining interpretability through their physics-based foundation [42–49]. For example, ref. [46] developed a hybrid diagnostic framework that couples artificial neural network modules with thermodynamic performance maps to infer compressor flow capacity reduction and turbine efficiency loss, enhancing fault detection accuracy and enabling earlier warnings.

Figure 1 illustrates a representative ANN-augmented thermodynamic model. Earlier frameworks [41] emphasized the integration of measurement data with thermodynamic models through calibration loops. In the present adaptation, an ANN correction module is added and trained on degradation scenarios to capture nonlinear effects, improving robustness under noisy or off-design conditions. Process map-based thermodynamic simulations are used to generate synthetic training data under both nominal and degraded states, supporting prediction of critical gas path parameters. Datasets representing degradation modes such as compressor fouling or turbine efficiency loss are created through cycle or component simulations, while ANNs are trained to map measurable engine parameters (pressures, temperatures, fuel flow) to unmeasured health indicators (flow capacity, efficiency loss, fault severity) [42–49]. By learning nonlinear relationships between observed sensor data and hidden degradation states, these hybrid models extend the capability of classical GPA and support robust, real-time health monitoring, with applications demonstrated in academic studies [42,43].

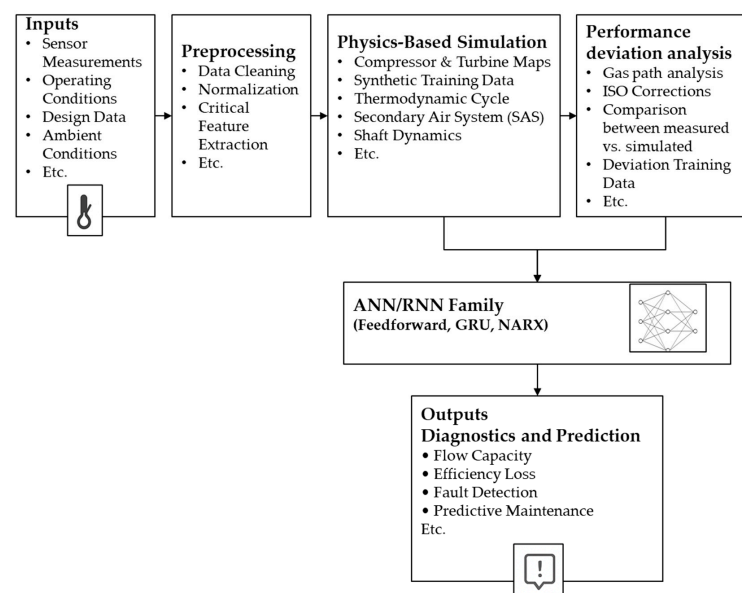


Figure 1. Process map diagram of a typical ANN-augmented thermodynamic model, modified from [41].

ANN-augmented thermodynamic models have demonstrated robustness to noisy sensor data and adaptability under off-design operating conditions [17,50–52]. They rely largely on synthetic datasets generated from thermodynamic simulations, preserve interpretability through their grounding in cycle physics, and focus on component-level health indicators. However, these models often require domain adaptation or retraining when applied to real engine environments, due to mismatches in operating conditions, sensor

noise, and unmodeled degradation effects [4,13]. Hybrid diagnostic frameworks have been developed that integrate ANNs with thermodynamic cycle simulations to estimate component degradation levels such as compressor flow capacity loss and turbine efficiency reduction, improving early fault detection using simulated deterioration data [42–45]. Distributed ANN-based gas path analysis (GPA) frameworks employing classification, auto-associative, and approximation networks have also been proposed to isolate and quantify multiple simultaneous faults, demonstrating robust identification, localization, and severity estimation under nonlinear and noisy conditions, and outperforming conventional GPA approaches [46–48,53].

2.1.1. Advantages of ANN-Augmented Thermodynamic Models

- Improved diagnostic accuracy compared with purely physics-based models, particularly in nonlinear or multivariate degradation scenarios [43,48];
- Fast inference speeds suitable for real-time and field-deployed applications. While inference is computationally efficient, the training phase can involve substantial overhead depending on the fidelity and validity of the underlying thermodynamic simulations [8,11];
- Ability to leverage synthetic training datasets even when historical fault data are limited [9,11];
- Demonstrated robustness to noisy sensor data and adaptability to off-design operating conditions [17,50–52].

2.1.2. Limitations of ANN-Augmented Thermodynamic Models

- Strong dependence on the quality and completeness of simulation data; any inaccuracies or biases in the input data directly propagate into the trained model, reducing its reliability in real-world applications [9,25];
- Limited coverage of degradation modes in the simulation dataset may bias predictions and reduce the model's ability to generalize to unobserved faults [9,25];
- Limited extrapolation beyond the design space represented in the simulation or training dataset [3,10];
- The “black-box” characteristics of ANNs can reduce physical interpretability unless explainable AI techniques are applied [17,50–53].

To address the former limitation, hybrid approaches increasingly incorporate physics-informed neural networks [41], grey-box models that combine thermodynamic principles with data-driven ANN modules [9], or explainable AI methods [54]. These strategies aim to ensure predictions remain physically meaningful while retaining the flexibility and adaptability of data-driven learning.

While ANN-augmented thermodynamic models provide valuable corrections to baseline physics, several challenges hinder their industrial deployment. First, models often require retraining or recalibration for each engine variant, as network weights are tuned to platform specific thermodynamic maps, limiting generalizability across fleets. Second, their accuracy depends strongly on the fidelity of the underlying simulation data; if the baseline thermodynamic model underestimates degradation effects, the ANN tends to propagate these biases. Third, performance outside the training domain, particularly during transient or abnormal conditions, remains unreliable, as most implementations focus on steady state diagnostics. In practice, maintaining accuracy outside the training domain (such as under load transients) often requires periodic recalibration or adaptive tuning [12].

2.2. Physics-Integrated Operational Architectures

Physics-integrated operational architectures combine real-time data with modular physical models and embedded AI elements such as physics-informed neural networks, ANN surrogates, and advanced optimization algorithms to deliver full system-level decision support and operational intelligence. These architectures extend beyond algorithm development, focusing instead on control, load allocation, preventive and predictive maintenance, and fleet-level operational planning [9,11,22,54,55].

Thermodynamic cycle models and component conservation equations provide the physically consistent backbone for these frameworks, ensuring accurate estimation of critical parameters such as turbine inlet temperature (TIT) and specific fuel consumption (SFC). Embedded AI modules enhance computational speed, improve robustness to sensor uncertainty, and capture nonlinear degradation effects that classical physics models alone cannot fully represent.

Figure 2 illustrates a representative architecture in which live measurements are fused with simulation models and AI surrogates to infer hidden states, optimize performance, and deliver actionable control recommendations. In this framework, gas-path sensor measurements are compared with model-based predictions, and the residual error is minimized to update estimated health parameters such as compressor flow capacity and gas turbine efficiency. This inverse diagnostic loop reflects gas path re-mapping approaches applied in industrial gas turbines, where [46] demonstrated its use for recalibrating component maps under degraded conditions to support operational diagnostics and performance monitoring. PINNs integrate physical constraints directly into their loss functions [19,54], while ANN surrogates trained on high-fidelity simulations replace computationally expensive physics modules to enable fast, real-time operation [11,55]. Optimization layers, such as particle swarm optimization (PSO), quadratic programming (QP) and other advanced techniques, are employed. In the case of PSO, this evolutionary algorithm, which is inspired by swarm behavior, efficiently explores the solution space, while QP provides a deterministic, mathematically rigorous approach for constrained optimization. Combined, these methods improve control performance and overall energy efficiency [22,54,55].

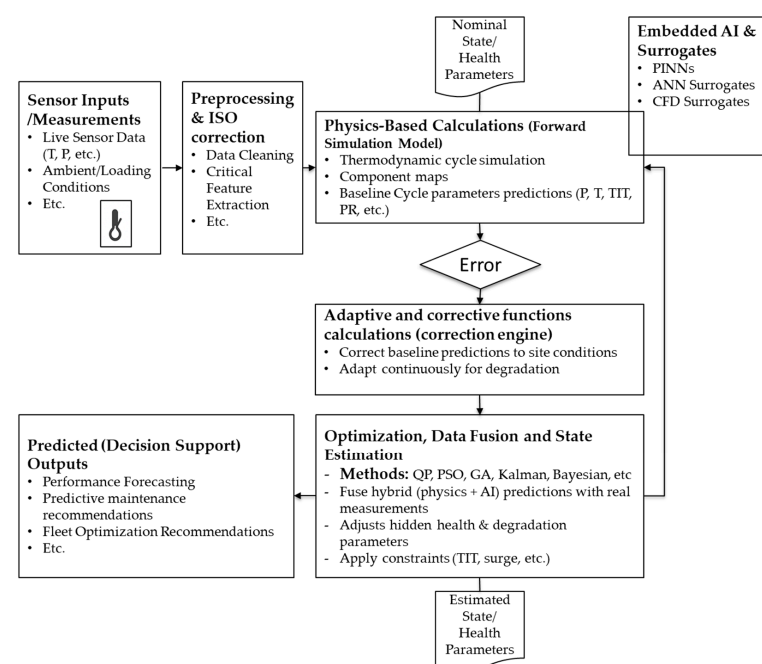


Figure 2. Process map diagram of a typical Physics-Integrated Operational Architecture, modified from [46].

A representative example of a physics-integrated operational architecture is the Electric Power Research Institute (EPRI) Turbine Digital Twin. This platform integrates live sensor and historian data with physics-based thermodynamic modules and AI-enabled virtual sensors to support performance forecasting, anomaly detection, and predictive maintenance [56]. EPRI has reported deployment across approximately 25 units spanning four turbine types, achieving data-to-calibrated-model turnaround times as short as one day under suitable infrastructure. Documented applications include trend monitoring of performance degradation, anomaly detection through virtual-sensor deviations, and short-term performance forecasting [56].

General Electric (GE) Vernova applies digital twin technologies via its SmartSignal predictive analytics platform, which reportedly combines physics-based performance baselines with artificial neural network (ANN)-based anomaly detection to identify deviations in real time. GE states that live data streams from turbine Supervisory Control and Data Acquisition (SCADA) systems and plant historian databases are compared continuously against model predictions, with residual monitoring enabling early detection of fouling, flow anomalies, or efficiency loss. SmartSignal is deployed across more than 7000 monitored assets worldwide, with GE reporting cumulative avoided losses exceeding USD 1.6 billion through reduced forced outages and optimized maintenance [57].

Siemens employs its Autonomous Turbine Operations and Maintenance (ATOM) digital twin framework, developed in collaboration with DecisionLab, to model fleet operations, maintenance workflows, supply chain logistics, and turbine behavior across the value chain [58]. Within this modular, agent-based simulation architecture, system components such as turbines, maintenance facilities, and supply chain entities are represented as interacting agents. The platform supports scenario testing, bottleneck analysis, and decision support for maintenance scheduling and investment planning. ATOM has been applied to Siemens Energy's aero-derivative industrial gas turbine SGT-A65, where it replaced legacy Excel-based forecasting tools and enabled more integrated, fleet-level operational planning [58].

In addition to industrial implementations, independent research and open-source frameworks have also played a central role in advancing physics-integrated operational architectures. For instance, the PySINDy package enables sparse identification of governing equations directly from measurement data [24], while operator-learning frameworks such as the Fourier Neural Operator (FNO) [28] provide resolution-invariant mappings of PDE systems. Deep-learning surrogates have also been demonstrated for modeling compressor manufacturing variations [59]. These academic and open-source contributions complement industrial platforms such as the EPRI Turbine Digital Twin [56], GE Vernova's SmartSignal [57], and Siemens' ATOM framework [58] by providing transparent, generalizable methods that can be validated, shared, and extended across research communities.

Table 1 provides a comparison across three major industrial digital twin platforms in terms of model structure and operational role. The EPRI Turbine Digital Twin integrates NPSS-based thermodynamic modules with AI-enabled virtual sensors to support operator-focused diagnostics, anomaly detection, and performance forecasting [56]. GE Vernova's SmartSignal combines artificial neural network (ANN)-based predictive analytics with thermodynamic baselines to enable predictive maintenance, downtime reduction, and fleet-wide operations and maintenance (O&M) optimization [57]. Siemens' ATOM framework employs an agent-based architecture that links physical models with AI optimization layers to coordinate load sharing, maintenance planning, and long-term fleet management [58]. These distinctions illustrate that EPRI emphasizes operator diagnostics, GE focuses on predictive maintenance, and Siemens prioritizes fleet-level optimization and coordination.

Table 1. Comparison of industrial digital twin platforms for gas turbines.

Platform	Model Structure	Operational Role
EPRI Turbine Digital Twin [56]	NPSS thermodynamic modules with AI-based virtual sensors	Operator-facing diagnostics, anomaly detection, and performance forecasting
GE Vernova SmartSignal [57]	ANN-based predictive analytics integrated with thermodynamic baselines	Predictive maintenance, downtime reduction, and O&M optimization across large fleets
Siemens ATOM [58]	Agent-based framework combining physical models with AI optimization layers	Fleet-wide coordination, load sharing, and predictive maintenance planning

Complementary research efforts extend these industrial implementations. For example, ref. [54] demonstrated a PINNs enabled intelligent digital twin that accurately predicted hidden variables such as turbine inlet temperature (TIT) and specific fuel consumption (SFC) under sensor uncertainty, while [55] implemented an artificial neural network (ANN) surrogate-based architecture for combined heat and power (CHP) gas turbines, enabling real-time modeling of degradation, load sharing, and control.

2.2.1. Advantages of Physics-Integrated Operational Architectures

- Real-time, system-wide representation of engine behavior using live sensor data integrated with physics-based thermodynamic cycle models such as NPSS [56];
- Embedded AI modules (ANN surrogates, PINNs, PcNNs) support adaptive performance modeling, virtual sensing, anomaly detection, and predictive maintenance [16,22,34,54,55];
- Demonstrated improvements in diagnostic accuracy and degradation tracking under nonlinear, noisy, or off-design operating conditions compared to classical gas path analysis [9,10,33];
- Computational efficiency through surrogate models (ANNs, operator-learning networks, CFD surrogates), enabling rapid virtual prototyping, design iteration, and real-time optimization [27,32,42];
- Scalable architectures suitable for plant-wide integration and fleet-level deployment, as demonstrated in EPRI's operator-focused twin, GE Vernova's SmartSignal platform, and Siemens' ATOM framework [56–58];
- Generative and model discovery methods (GANs, RNNs, SINDy) augment sparse datasets and extract interpretable governing equations, enhancing model robustness for rare-event prognostics and transient validation [39,45].

2.2.2. Limitations of Physics-Integrated Operational Architectures

- Integration complexity, including sensor synchronization, model calibration, and the need for robust data pipelines when interfacing with existing Condition Monitoring Systems (CMS) and plant control networks [56];
- Dependence on the quality and coverage of training data; poor representation of degraded or transient conditions may bias predictions [9,42,49];
- ANN-heavy approaches exhibit “black box” behavior with reduced interpretability unless constrained by physics-informed or explainable AI methods [42,47,51];
- High computational costs during training for PcNNs, PINNs, and generative surrogates due to embedded PDE residuals and automatic differentiation, with additional challenges in stability and convergence [20,21,29];
- Cross-platform generalization is limited; calibrated twins often require domain adaptation, and standardized benchmarks for validation are lacking [42,49,55];

- Cybersecurity and latency concerns in interconnected SCADA and historian environments, exposing vulnerabilities in critical infrastructure [56,57];
- High development and lifecycle costs, requiring multidisciplinary expertise in thermodynamics, control systems, data science, and IT infrastructure [16,22,54].

While PcNNs and CFD surrogates demonstrate strong performance in simulations, their validation against experimental data remains challenging. Sparse or noisy sensor measurements can limit the accuracy of reconstructions, particularly for flow or thermal fields where ground-truth data are difficult to obtain. Ref. [21] noted that PINN-based reconstructions of turbine passage velocity and temperature fields were constrained by the limited availability of high-fidelity sensor data, reducing generalizability in actual gas turbines. Similarly, ref. [39] emphasized that physics-informed heat transfer models require carefully designed experimental benchmarks to ensure reliable predictions under high-temperature operating conditions. These challenges highlight that robust experimental validation is a critical step before PcNNs can be confidently adopted for industrial gas turbine applications.

2.3. Physics-Constrained Neural Networks and CFD Surrogates

Physics-constrained neural networks embed governing physical laws such as Navier–Stokes, continuity, and energy balance equations directly into the training process, ensuring physically consistent predictions even when labeled data are sparse or noisy [8,10,21,23,31,60]. In this framework, geometry, boundary conditions, and operating parameters are provided as inputs, while conservation laws are enforced as embedded constraints to maintain physical consistency. These constraints may be applied as soft penalties in the loss function or as hard constraints within the network architecture, thereby ensuring adherence to principles such as the non-isothermal Navier–Stokes equations [26] and energy balance relations. By integrating such physical priors, PcNNs reduce dependence on dense experimental datasets and bridge the gap between purely data-driven models and high-fidelity physics-based simulations. A typical PcNNs workflow and computational fluid dynamics (CFD) surrogate pipeline for gas turbine applications is shown in Figure 3, where geometry, boundary conditions, and operating parameters serve as inputs, and embedded constraints ensure flow, thermal, and structural predictions remain consistent with fundamental physics.

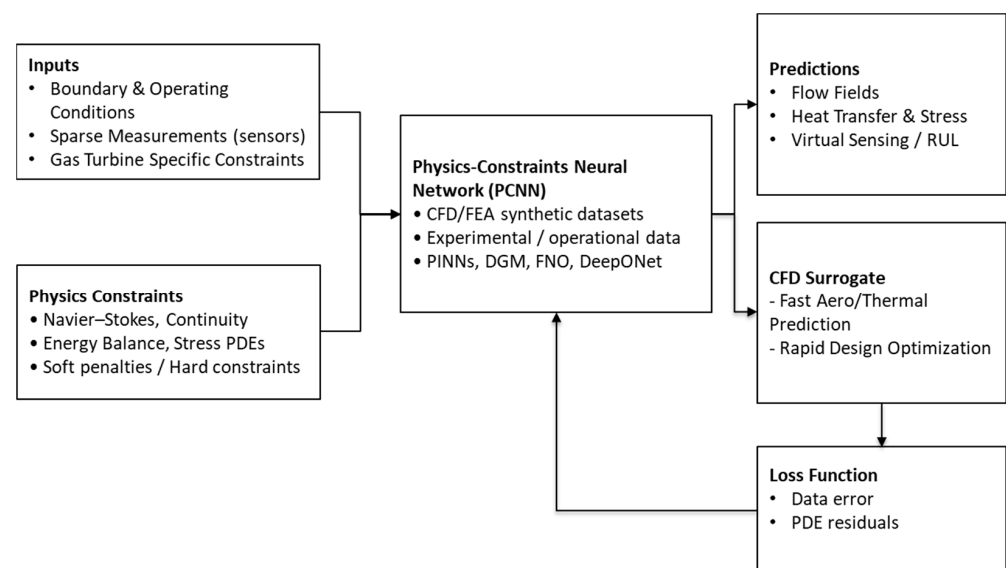


Figure 3. Process map diagram of a typical physics-constrained neural networks, modified from [31].

Applications in gas turbines include flow-field inference, heat-transfer analysis, structural stress estimation, and virtual sensing. For example [19], applied Physics-Informed Neural Networks (PINNs) to reconstruct temperature and velocity fields within turbine passages using sparse sensor measurements, embedding Navier–Stokes and energy equation residuals into the loss function to improve generalization and physical fidelity. Ref. [21] reviewed broader PINN applications in industrial gas turbines, including aerodynamic and aero-acoustic field estimation, blade flutter and fatigue prediction, combustion instability diagnostics, and turbine vane heat-transfer modeling. Likewise, ref. [39] demonstrated how physics-informed learning can solve conductive and convective heat-transfer PDEs in high-temperature engineering applications, showing that embedded physics constraints can yield reliable predictions even when experimental thermal data are limited. Foundational approaches include PINNs [31] and the Deep Galerkin Method (DGM) [32], which enable solving forward and inverse problems for nonlinear PDEs without requiring structured meshes. These examples highlight the suitability of PcNNs for predicting coupled aero-thermal behavior in turbine components, where flow and heat transfer are closely linked to degradation mechanisms such as thermal fatigue and creep. PcNNs also enable the development of CFD surrogate models, significantly reducing computational cost compared to conventional solvers. Ref. [61] introduced a transformer-based surrogate for three-dimensional compressible flow prediction in axial compressors without mesh generation, enabling near-instantaneous aerodynamic predictions for early-stage design optimization. Ref. [59] applied deep learning to model the impact of manufacturing and build variations on multistage axial compressor aerodynamics, showing how high-fidelity synthetic datasets can improve surrogate accuracy for complex multistage flows. Other surrogate models leveraging convolutional neural networks, graph neural networks, and generative models trained on high-fidelity CFD datasets [1,13,17,29,62,63] have achieved significant computational speed-ups while maintaining accuracy. Ref. [45] developed a transformer-based, data-driven AI model for turbomachinery compressor aerodynamics that enables rapid approximation of 3D flow solutions from CFD data. Although accuracy decreases for designs far outside the training distribution, the method provides a computationally efficient surrogate for aerodynamic design and optimization. Trained on high-resolution compressor flow simulations, the model reproduces key aerodynamic features and allows derivation of performance metrics such as efficiency. Operator learning architectures extend these capabilities further: the Fourier Neural Operator (FNO) and Fourier DeepONet can learn full solution operators for entire families of PDEs, enabling mesh-independent, resolution-invariant predictions across different geometries and operating regimes [25,28,64,65]. These approaches support virtual sensing of otherwise unmeasurable states, accelerated aerodynamic and thermal design iterations, and physically consistent diagnostics and prognostics.

2.3.1. Advantages of Physics-Constrained Neural Networks and CFD Surrogates

- Physically consistent outputs even when labeled data are sparse or noisy [13,22,61];
- Reduced reliance on labeled datasets through embedded physical laws [3,8,10,21,23,60];
- Fast surrogate predictions for flow, thermal, and stress analyses, enabling rapid design optimization and diagnostics [1,13,17,29,61–63];
- Resolution-invariant learning using operator learning frameworks Fourier Neural Operator (FNO), Fourier DeepONet for generalization across geometries and operating conditions [25,28,64–68].

2.3.2. Limitations of Physics-Constrained Neural Networks and CFD Surrogates

- High computational cost during training due to embedded PDE residuals and automatic differentiation [10,23];
- Numerical stability and convergence challenges, especially in multi-physics environments [13,62,63];
- Limited transferability across turbine platforms unless domain adaptation or transfer learning is applied [1,17,29];
- Lack of standardized benchmarks and evaluation protocols for cross-platform validation [8,21].

Overall, while PcNNs and CFD surrogates provide a powerful bridge between data-driven models and physics-based solvers, their adoption is constrained by several unresolved challenges. Training remains computationally intensive due to embedded PDE residuals and automatic differentiation, and convergence is often unstable in multi-physics environments [10,23,62,63]. The lack of standardized benchmarks and validation datasets across turbine platforms [8,21] hinders reproducibility and consistent evaluation, while limited transferability means models often require costly retraining when applied to different engine types [1,17,29]. These issues highlight that further progress will depend on improved numerical schemes, domain-adaptive learning, and benchmark initiatives that can establish reliability across operating regimes and gas turbine platforms.

2.4. Generative and Model Discovery Approaches

Generative and model discovery approaches address challenges of data scarcity, interpretability, and modeling of complex multicomponent interactions in gas turbine systems. These methods are particularly valuable when operational fault data are rare, extreme events are underrepresented, or physics-based models are incomplete or computationally expensive for real-time deployment. Generative models such as generative adversarial networks (GANs), variational autoencoders (VAEs), and recurrent neural networks (RNNs) have been applied to create synthetic operating scenarios, expanding limited datasets and improving robustness under rare or transient conditions [8,39,40]. Model discovery techniques, including sparse identification of nonlinear dynamics (SINDy) and its PySINDy implementation, extract simplified governing equations directly from data to support reduced-order modeling and interpretable control [24,30,69].

2.4.1. Generative Models

GANs, VAEs, and RNNs generate synthetic sensor data representing rare or extreme scenarios, enabling robust model training under limited data conditions [8]. Beyond diagnostics, generative AI has been applied to engineering design, where neural network-aided approaches systematically explore design spaces, integrate multidisciplinary objectives, and accelerate concept evaluation. For example, one study demonstrated how a neural network-based technological evolution system optimized product development by learning from prior designs [48]. Similar methodologies have been used for circular design [66], process control and fault diagnosis [67], and multidisciplinary product specification [68]. In gas turbine contexts, such generative frameworks improve rare-event coverage, enhance design robustness, and reduce validation costs. RNN architectures have further been shown to deliver super real-time transient thermal predictions in convection systems, demonstrating potential for virtual sensing and rare-event prognostics [40].

2.4.2. Model Discovery

Model discovery methods complement data generation by extracting parsimonious governing equations directly from data. Approaches such as SINDy [69] and PySINDy [24]

produce interpretable reduced-order models suitable for control, fault detection, and diagnostics in regimes not captured by classical simulations [30]. These techniques can be combined with synthetic data generation to expand scarce datasets and extract physically meaningful dynamics. For gas turbine applications, such frameworks support robust diagnostics and prognostics under limited fault data, with validation strategies that include cross-checking synthetic scenarios against high-fidelity CFD or thermodynamic simulations. Recent advances, such as physics-informed diffusion models for anomaly detection [70], highlight the potential of combining generative AI with physics-based constraints to ensure realistic and trustworthy predictions in intelligent digital twins.

However, applying model discovery in practice requires careful feature selection, noise reduction, and domain expertise to prevent spurious dynamics from being misidentified as governing equations. Figure 4 illustrates this combined workflow: generative methods expand scenario coverage, while model discovery techniques extract simplified yet physically consistent system dynamics. This integration improves rare-event representation and strengthens hybrid intelligent digital twins.

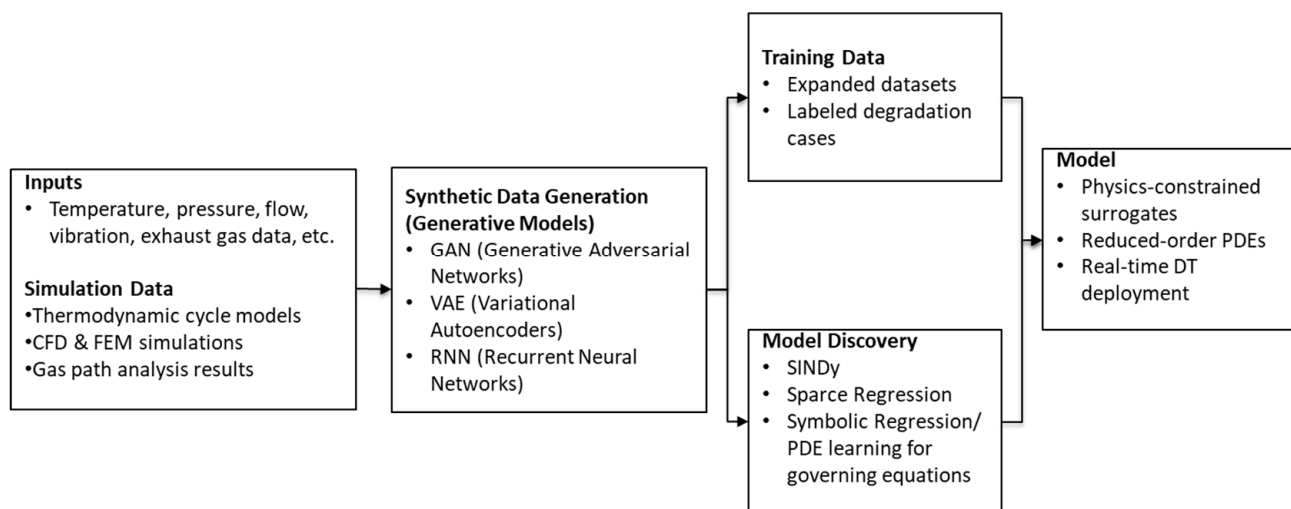


Figure 4. Process map diagram of a typical Generative and Model Discovery Approaches, modified from [24,69].

2.4.3. Emerging Architectures

Recent advances extend these methods to scalable multi-physics inference. Physics-informed graph neural Galerkin networks [36], graph convolutional neural networks [37], and Discretization Net [27] embed mesh connectivity and discretization principles into learning architectures, achieving high accuracy on irregular domains, accelerating Navier–Stokes solutions, and improving virtual sensing from sparse measurements [30]. Energy-preserving designs such as Lagrangian Neural Networks (LNNs) [33] further enable robust inference across interdependent subsystems. Together, these advances move beyond data augmentation and equation discovery toward physics-consistent digital twins, with applications in hot-gas-path monitoring, secondary air system modeling, and adaptive flow/thermal field reconstruction.

2.4.4. Advantages of Generative and Model Discovery Approaches

- Synthetic data augmentation, improving model robustness under rare or previously unobserved conditions [8,40];
- Equation discovery and interpretability, supporting reduced-order modeling, control and diagnostics [24,69];

- Scalable graph-based and energy-preserving architectures for complex multi-component systems [30–37].

2.4.5. Limitations of Generative and Model Discovery Approaches

- Generative models may produce physically unrealistic signals if not properly constrained [8,40];
- Equation discovery methods such as SINDy are sensitive to noise and feature selection [69];
- Graph-based and energy-preserving neural architectures can be computationally demanding when scaled to full-system or fleet-level applications [27,30,36,37].

Integrating generative, discovery, and graph-based methods into industrial digital twins requires rigorous validation to ensure regulatory compliance, user confidence, and robustness under unseen operating conditions. Without explicit physical constraints, generative models may produce infeasible outputs such as negative mass flow rates, unrealistic efficiency values, or fail to capture the coupled thermo fluid structural dynamics of gas turbines. Similarly, model discovery techniques risk misidentifying governing equations when trained on noisy or incomplete datasets. To ensure reliability, validation mechanisms are essential. Strategies include embedding physics consistency checks (e.g., conservation of mass and energy laws) into training pipelines, applying adversarial filtering to reject nonphysical synthetic signals, and cross validating synthetic scenarios against high fidelity CFD or thermodynamic simulations before deployment [8,69]. Comparable safeguards are already being adopted in other domains: in aerospace, physics informed GANs generate turbulence consistent flow fields, while in manufacturing, VAEs constrained by metallurgical rules ensure plausible microstructural evolution [8,39,40]. These parallels underscore that combining generative AI with physics-based validation is essential to ensure synthetic data strengthens rather than undermines the robustness of intelligent gas turbine digital twins.

3. Results

The comparative analysis shows that no single hybrid AI method is best for all gas turbine digital twins. Each approach works better in certain contexts. The choice depends on the goal such as online diagnostics, component-level optimization, or fleet-wide asset management. Results also depend strongly on available computational resources and the quality of data. To evaluate these approaches systematically, a novel maturity classification framework is developed based on five key criteria:

- Data dependency measures how strongly a method relies on large datasets versus embedding physical constraints to reduce data requirements [29,42,53].
- Physical interpretability evaluates whether model outputs are explainable and grounded in physical laws or remain black-box predictions [31,60,64].
- Deployment complexity reflects the infrastructure and resources needed for implementation, ranging from minimal calibration to full HPC-enabled workflows [34,56].
- Workflow compatibility considers the ability to integrate with existing thermodynamic, CFD, or SCADA-based systems, a critical factor for operational adoption [35,57,66].
- Real-time capability assesses whether methods are limited to offline analysis, partially validated for online use, or demonstrated in real-time diagnostics and control [9,16,56].

In this framework, maturity is assessed relatively across methods, with supporting references for each category. Scores of 1–2 indicate low maturity (experimental methods with limited applicability), 3 indicates medium maturity (methods with potential but still facing technical or validation gaps), and 4–5 indicate high maturity (robust, validated, and scalable methods suitable for deployment). Table 2 presents the scoring rubric used to

evaluate hybrid AI methods for intelligent gas turbine digital twins across five criteria: data dependency, physical interpretability, deployment complexity, workflow compatibility, and real-time capability. Maturity scores were assigned using this rubric together with the advantages and limitations of each method discussed in Section 2, ensuring the assessment reflects both theoretical capabilities and practical constraints. This provides a transparent, evidence-based foundation for the comparative maturity framework in Figure 5, where a radar plot illustrates performance across the five criteria on a 1–5 scale.

Table 2. Scoring rubric for maturity assessment of hybrid AI methods, with supporting references.

Criterion	Score 1–2 (Low)	Score 3 (Medium)	Score 4–5 (High)	Key References
Data Dependency	Strong reliance on large datasets; limited generalization; often restricted to synthetic	Moderate reliance; able to use a mix of simulation and sensor	Low reliance; physics constraints or transfer learning reduce data requirements	[29,42,53]
Physical Interpretability	Black-box predictions with little or no physical grounding	Partial transparency through hybrid coupling or explainable AI techniques	High interpretability with embedded governing equations or physics-preserving networks	[31,60,64]
Deployment Complexity	Simple deployment; minimal infrastructure needs	Moderate effort; requires calibration, pipelines, or custom tuning	High complexity; requires HPC, robust pipelines, and/or cybersecurity layers	[34,56]
Workflow Compatibility	Standalone; minimal integration with thermodynamic or CFD models	Partial integration with selected simulation modules	Full compatibility with SCADA, NPSS, CFD, and fleet-level digital twin workflows	[35,57,66]
Real-Time Capability	Limited to offline or retrospective analysis	Medium suitability; fast inference but not yet fully validated online	Demonstrated real-time diagnostics, control, and predictive maintenance	[9,16,56]

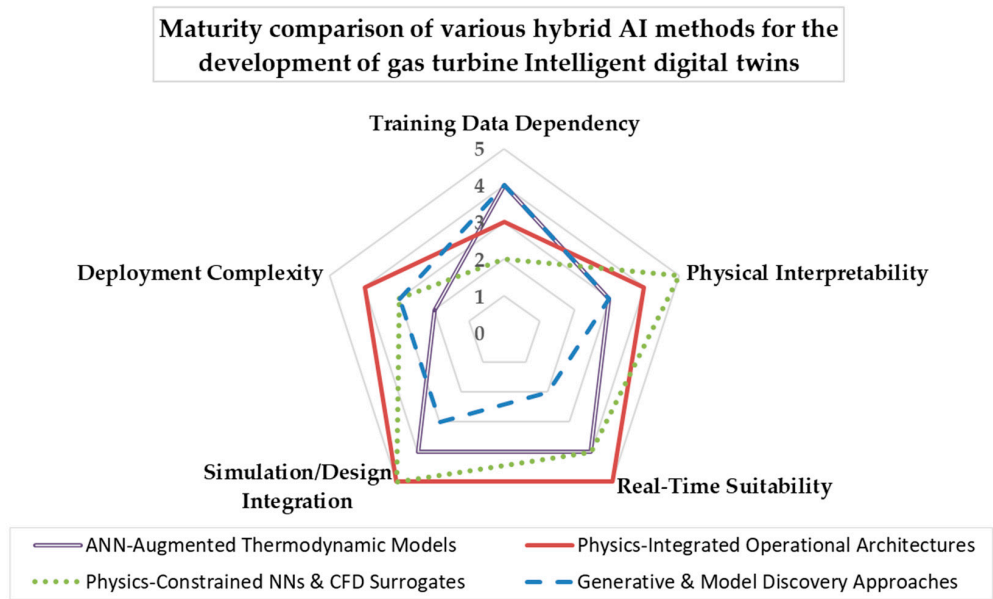


Figure 5. Radar Plot of Hybrid AI Methods Maturity for Gas Turbines Intelligent Digital Twins.

Table 3 presents a side-by-side synthesis of each category, highlighting typical advantages, limitations, and data/computational requirements. This tabular summary adds

context to the maturity framework by clarifying, for example, that ANN-augmented thermodynamic models offer rapid online inference but depend on baseline fidelity, while physics-constrained neural networks provide higher interpretability at the cost of greater training complexity. Similarly, physics-integrated operational architectures are scalable to fleet-level deployment but face integration and cybersecurity challenges, whereas generative and model-discovery methods remain data-intensive and at an earlier stage of industrial maturity.

Table 3. Hybrid AI categories—advantages, limitations, and typical data/computational requirements.

Category	Advantages	Limitations	Typical Use Cases
ANN-Augmented Thermodynamic Models	Improves diagnostic sensitivity; compensates for degradation drift; fast inference for online use	Limited extrapolation outside training domain; dependent on baseline model fidelity	Moderate data needs (sensor + baseline cycle data); low–moderate compute (training offline, lightweight online inference)
Physics-integrated operational architectures	Maintains physical consistency; integrates sensor data with fleet/asset models; scalable to plant operations	Calibration effort; higher deployment complexity; potential cybersecurity concerns	Large operational datasets (SCADA, historian logs); moderate–high compute (integration, pipelines, dashboards)
Physics-constrained neural networks and CFD surrogates	High interpretability; preserves conservation laws; enables rapid flow-field predictions	Training complexity; surrogate validation required; performance tied to CFD reference quality	High-fidelity CFD or experimental datasets; high compute for training; reduced cost at inference
Generative and model discovery approaches	Identifies governing equations directly from data; adaptable to unseen conditions; supports synthetic data generation	Data hungry; interpretability varies; early-stage maturity for industrial adoption	Large, diverse datasets (including synthetic/augmented); high compute for training (GANs, VAEs, SINDy), variable at inference

First, ANN-augmented thermodynamic models are entry-level solutions. They require substantial amounts of synthetic or historical datasets [10,42]. Their interpretability is moderate, as they retain some transparency from the thermodynamic baseline but depend on black-box ANN layers. Deployment complexity is relatively low, which makes them practical for component-level diagnostics and degradation tracking. They integrate reasonably well with simulation workflows such as gas path analysis and provide medium–high real-time suitability, making them effective for rapid health monitoring [10,41,43,71,72].

Second, physics-integrated operational architectures represent the most mature category. They combine live sensor data with modular physics-based models, AI surrogates, and optimization algorithms [56–58]. Their interpretability is high because the physics modules anchor the AI components, and they integrate strongly with simulation and design tools. These systems achieve the highest maturity for real-time suitability, enabling adaptive control, predictive maintenance, and fleet-level decision support. However, they require substantial volumes of operational data and present high deployment complexity, including the need for robust data pipelines and advanced computing infrastructure [56].

Third, physics-constrained neural networks and CFD surrogates provide a balanced trade-off. By embedding governing equations directly into training, they reduce data dependency and achieve very high interpretability [17,21,29,31]. They are particularly suitable for virtual sensing, thermal and flow field reconstruction, and rapid design iteration.

Their strong alignment with CFD and structural analysis workflows ensures excellent simulation compatibility, while their real-time suitability is medium to high once training is complete. Deployment complexity is moderate, reflecting the significant training demands and computational cost [25,36,62].

Finally, generative and model discovery approaches remain at an early stage of maturity. Generative models such as GANs, VAEs, and RNNs can create synthetic datasets to address data scarcity, while discovery methods such as SINDy extract simplified governing equations [8,24,69]. These approaches show promise for rare-event prognostics, transient validation, and synthetic data augmentation, but their interpretability is only moderate and their reliance on diverse datasets remains high [8,39,40]. Real-time suitability is limited, as most applications remain experimental or offline. Their simulation and design integration are also low-to-moderate, since they typically complement rather than directly embed within engineering workflows. Deployment complexity is moderate, but validation requirements are significant, as unconstrained generative models may produce non-physical signals. Overall, these methods should be regarded as exploratory rather than production-ready, representing a frontier for future research rather than a proven industrial standard.

Together, these categories show a clear path of development. It begins with ANN models for component-level monitoring, moves through physics-constrained surrogates and mature hybrid operational architectures, and extends to generative and discovery methods. These newer approaches are still in the research stage, but they offer ways to handle rare events and data gaps. This progression reflects an evolution toward fully adaptive, fleet-integrated intelligent digital twins.

4. Discussion and Future Trends

This section examines the maturity of hybrid AI methods, the barriers that limit their deployment, and the opportunities for advancing intelligent gas turbine digital twins. It also highlights practical integration challenges and proposes a layered hybrid AI framework for future development.

4.1. Comparative Maturity of Hybrid AI Methods

The maturity comparison presented in the results Section 3 shows a clear progression from component-level entry points to fully integrated, adaptive system-level platforms. ANN-augmented thermodynamic models remain practical early-stage solutions due to their low deployment complexity and ease of implementation. However, their heavy reliance on synthetic datasets and limited interpretability restricts their adaptability to evolving operational scenarios [47,49]. At the opposite end, hybrid intelligent digital twins with embedded AI represent the highest maturity level, enabling real-time adaptive control, predictive maintenance, and fleet-wide optimization through continuous sensor data integration [5,11,21,50,56]. These advanced capabilities, however, come at the cost of high deployment complexity, requiring significant infrastructure and robust data pipelines.

Physics-constrained neural networks and CFD surrogates provide a middle ground. By embedding governing equations, they reduce data needs and improve interpretability. This makes them well suited for virtual sensing, thermal and flow field reconstruction, and rapid design iteration [17,25,27,28,31,33–36,39,45,60].

Generative and model discovery approaches remain less mature. They address rare-event data scarcity and improve interpretability by generating synthetic scenarios and discovering governing equations [8,24,30,37,40,69]. However, most applications are still offline, with limited real-time use. Similar patterns are visible in aerospace, where physics-informed operators, neural fields, and graph-based surrogates provide scalable alternatives

to high-fidelity CFD for aerodynamic flow prediction and aeroelastic analysis [73–76]. These advances show lessons that can transfer to gas turbine digital twins, which face similar challenges of nonlinear dynamics, multi-physics coupling, and real-time requirements.

Overall, the trajectory is clear: starting with ANN-based models, moving through physics-constrained surrogates and hybrid operational frameworks, and extending to generative approaches for rare-event readiness. This aligns with evidence that hybrid AI outperforms purely data-driven or purely physics-based methods under degraded or uncertain conditions [19,22,42,43]. It also reflects industry trends toward multi-fidelity data fusion [72] and Bayesian PINNs for uncertainty-aware decision-making [71].

4.2. Challenges and Opportunities

Despite the clear progress across hybrid AI categories, several common challenges continue to limit their full-scale deployment in intelligent gas turbine digital twins. Data scarcity remains a primary barrier: high-quality degraded-condition datasets are rare, and there is no standardized, open benchmarking framework for training and validation [3,4,8,11,62], making it difficult to evaluate competing approaches or accelerate technology adoption. High computational cost is another constraint, as real-time CFD surrogates and physics-constrained networks require substantial processing resources to operate within industrial timeframes [25,27,28]. Integration complexity adds further difficulty. Embedding hybrid AI into existing SCADA and plant control networks demands secure, low-latency data pipelines and strict compatibility with established operational protocols [56]. Integration must not compromise these core functions.

At the same time, the opportunities are significant. Generative models can expand limited datasets with realistic rare-event scenarios, improving model robustness under critical fault conditions [8]. Transfer learning offers a pathway to cross-platform generalization, enabling models trained on one turbine type to adapt efficiently to others [9]. Multimodal sensor fusion can strengthen situational awareness by integrating vibration, temperature, acoustic, and operational logs into a unified operational picture [72]. Embedding hybrid AI into secure, cloud-enabled digital twin platforms would support fleet-level lifecycle optimization, centralized monitoring, and coordinated decision-making across assets [56–58]. Finally, uncertainty quantification methods such as Bayesian PINNs can improve the trustworthiness of predictions, providing operators with confidence in AI-driven recommendations under uncertain or degraded conditions [71].

4.2.1. Operational Integration of Hybrid AI Intelligent Digital Twins

A critical but often underexplored challenge for hybrid AI digital twins is their integration into the operational ecosystem of gas turbine plants. While SCADA and cybersecurity concerns are frequently mentioned, successful deployment also depends on alignment with control room practices, reliability standards, and regulatory compliance.

First, control room practices and operator trust are decisive for adoption. Hybrid AI predictions must be communicated through operator-facing dashboards in terms of conventional key performance indicators (e.g., efficiency, heat rate, turbine inlet temperature) rather than opaque anomaly scores. Experience from the Electric Power Research Institute (EPRI) Turbine Digital Twin shows that embedding AI-driven virtual sensors and predictive health indicators directly into SCADA displays improves operator acceptance and supports daily decision-making [56]. Similarly, GE Vernova's digital twin integrates AI outputs into performance dashboards that enable load adjustments and cost-optimization strategies [57]. Siemens' ATOM platform extends this approach to fleet operations, combining AI with physical models for load sharing and maintenance planning [58]. These examples demonstrate that human machine interface considerations are as critical as modeling accuracy.

Second, reliability standards and safety protocols define the operational envelope within which hybrid AI must function. Gas turbine fleets operate under strict reliability and safety requirements, with condition-based maintenance strategies ensuring that inspection intervals and load allocations do not compromise safety margins. Hybrid AI systems must therefore embed physical constraints into optimization loops to guarantee that recommendations remain within certified operational limits. Previous reviews of diagnostic and monitoring practices [6,41,62] emphasize that long-term adoption of hybrid AI depends on demonstrable improvements in reliability and compliance with established maintenance frameworks.

Third, cybersecurity and data pipeline integrity impose additional restrictions on integration architectures. Since SCADA systems are designated as critical infrastructure, hybrid AI models must be deployed within secure, low-latency data pipelines that preserve both availability and resilience. Ref. [11] highlighted that predictive maintenance platforms must address issues of sensor synchronization, robust pipelines, and secure integration with existing supervisory control systems. Likewise, ref. [51] emphasized that embedding advanced ML frameworks into physical systems introduces additional complexity in ensuring stable and secure operation. These studies underscore that operational integration is not only a technical challenge but also a regulatory and organizational one.

Together, these considerations highlight that hybrid AI deployment requires not only algorithmic advances but also the design of secure, operator-trusted, standards-compliant integration pathways that enable intelligent digital twins to be adopted at scale in critical energy infrastructure.

4.2.2. Cybersecurity and Integration Challenges in Hybrid Digital Twins

A critical challenge for hybrid digital twins lies in resolving technical bottlenecks related to cybersecurity, data integrity, and real-time responsiveness. Practical engineering solutions from industrial deployments and recent research provide a roadmap for robust integration.

Edge-cloud collaborative computing distributes tasks between local and remote resources. Edge modules collocated with SCADA systems perform lightweight anomaly detection and preprocessing to reduce latency and ensure resilience during communication loss, while cloud platforms handle fleet-wide optimization and long-horizon prognostics [34,58,73].

Fault-tolerant data pipelines are equally important, since continuous sensor ingestion must be safeguarded against packet loss, corruption, or tampering. Modern architectures employ redundant streaming protocols (e.g., Kafka, MQTT), multilayer verification, and anomaly filters to prevent corrupted data from propagating through diagnostic workflows [2,6,74]. Comparative evaluations of digital twin deployments confirm that resilience to cyber-physical disruptions is essential for certification and compliance [73].

Embedding physical constraints in control loops prevents unsafe or nonphysical recommendations from purely data-driven models. Hybrid controllers enforce thermodynamic balances and gas-path invariants, while physics-informed neural networks (PINNs) embed conservation laws directly in the loss function [10,60]. Such physics-constrained loops improve interpretability, reduce extrapolation risks, and increase regulatory acceptance [73].

Real-time inference acceleration is required to meet SCADA cycle demands, since sub-second responses are often beyond the reach of CFD or other high-fidelity models. Surrogate approaches, such as transformer-based and deep-learning architectures, when accelerated using GPUs, FPGAs, and model compression methods (e.g., pruning and quantization), can achieve the necessary real-time performance. Demonstrations confirm that

surrogate-assisted cycle modeling can scale to plant-level digital twins with significant speed gains [32,63,76]. Large-scale graph-based surrogate frameworks further extend this capability, enabling rapid 3D flow field prediction and scalable aerodynamic simulations [74,75].

Cybersecurity hardening remains essential, as SCADA networks are vulnerable to false-data injection and adversarial manipulation. Embedding physics-based residual checks and NPSS-derived baselines enables detection of inconsistencies that may signal tampering [56]. More advanced safeguards include physics-informed GAN filters, adversarial validation strategies, and zero-trust authentication protocols.

Recent studies further extend these strategies. Surrogate-based methods and graph machine learning approaches demonstrate scalability in aerodynamic and turbomachinery simulations [73–76], while physics-informed diffusion models illustrate how generative approaches can be constrained to detect subtle or adversarial anomalies in complex multivariate sensor environments [70].

Together, these engineering solutions show that addressing cybersecurity, data integrity, and responsiveness requires not only algorithmic advances but also layered verification, distributed computing, and physics-informed safeguards to enable secure and scalable deployment of hybrid digital twins in critical energy infrastructure [70,73–76].

4.3. Future Research Directions and Proposed Hybrid AI Framework

Future research in hybrid AI for intelligent gas turbine digital twins should focus on four complementary priorities. First, developing standardized benchmarks and open datasets is essential to accelerate model innovation and ensure fair, transparent performance comparisons across hybrid AI methodologies [3,4,8,11,62]. Second, advancing uncertainty quantification and physics-informed transfer learning offers a pathway to more robust and generalizable intelligent digital twin solutions, capable of adapting across diverse turbine platforms and operating conditions [9,19,71]. Third, the industry must establish secure, scalable integration strategies that support plant-wide intelligent digital twins with real-time monitoring and predictive control, while ensuring cyber resilience and operational reliability [11,51,56–58]. Finally, there is significant potential in ensemble and layered hybrid AI architectures that merge ANN-based diagnostics, physics-constrained surrogates, and generative models to deliver rare-event readiness, improved interpretability, and faster deployment timelines [17,22,24,30,47]. To address these priorities, this review presents a layered hybrid AI framework for gas turbine intelligent digital twins. The framework shown in Figure 6 is organized into four synergistic layers: a Physics Backbone anchoring thermodynamic and component models, an AI Modeling Layer integrating advanced learning architectures, a Robustness and Uncertainty Layer incorporating generative models and uncertainty quantification, and an Optimization and Intelligence Layer enabling predictive control and fleet-level decision-making. Elements of this framework are already reflected in industrial practice. The EPRI Turbine Digital Twin integrates NPSS-based thermodynamic modules with AI-enabled virtual sensors for parameter inference, exemplifying the physics backbone and AI modeling layers [56]. Siemens' ATOM platform demonstrates distributed optimization for fleet operations [58], while GE Vernova's SmartSignal system operationalizes predictive analytics across large asset portfolios [57]. These case studies confirm that key components of the proposed architecture are being deployed, underscoring its practical relevance for intelligent digital twins.

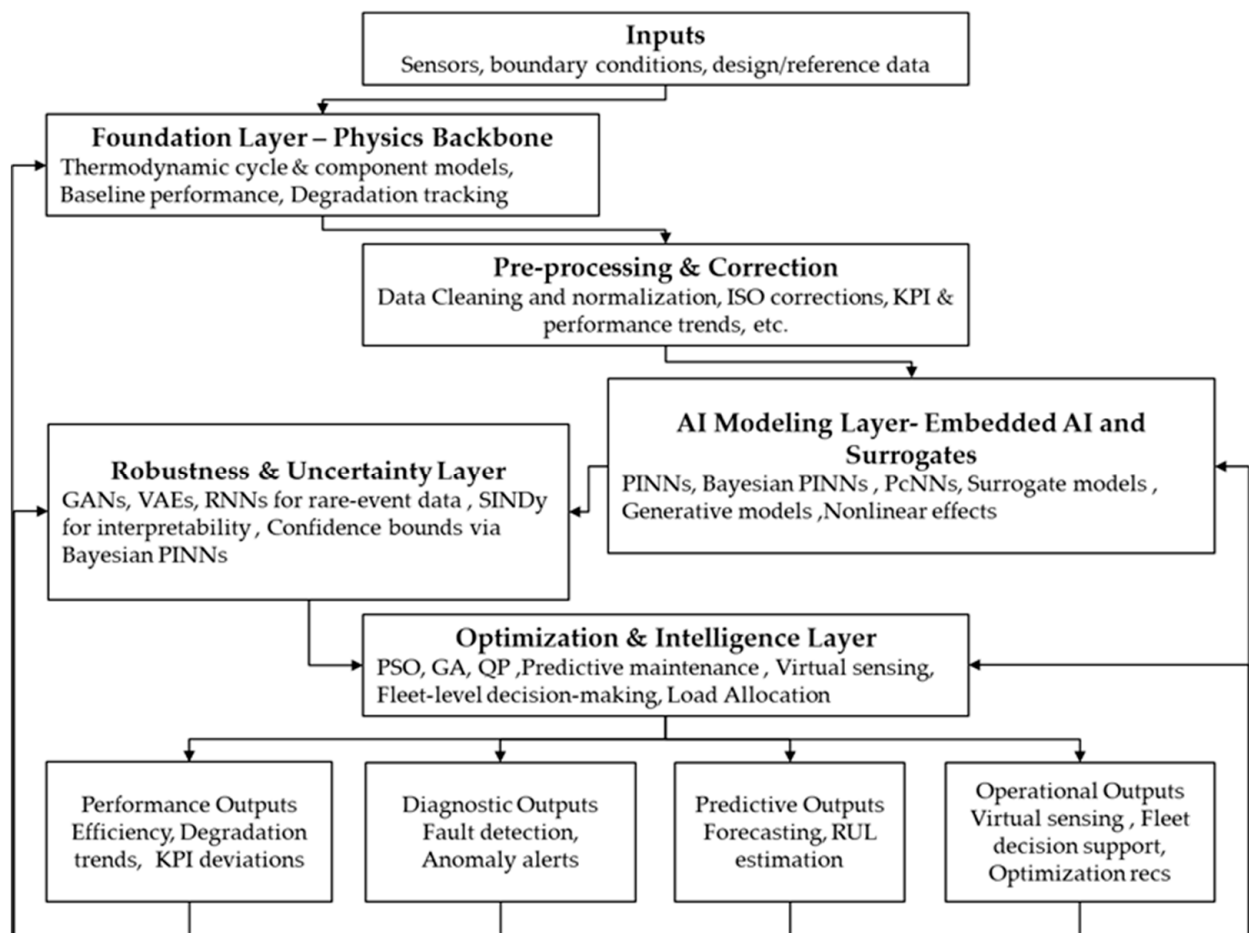


Figure 6. Proposed layered hybrid AI framework for gas turbine intelligent digital twins, modified in part from industry cases [56–58].

4.3.1. Physics Backbone (Foundation Layer)

At the base, physics-based thermodynamic and component models provide consistent performance predictions and interpretable degradation characterization. This grounding in conservation laws and engineering principles ensures trustworthy baseline estimates for efficiencies, flow capacity, and heat balance residuals. In addition, vibration channels from axial and radial casing probes supply structural and rotor dynamic health indicators. These raw time-domain signals form a complementary stream that anchors the twin not only in thermodynamic conservation but also in vibration behavior, enabling a more comprehensive baseline of turbine condition.

4.3.2. AI Modeling Layer

Building on the physics foundation, machine learning modules extend model fidelity and speed. Physics-informed neural networks [31] and their Bayesian or variational extensions [29,71] enable uncertainty-aware diagnostics. Operator-learning surrogates such as long short-term memory networks, graph neural networks [34,35], and neural operators—including the Fourier neural operator and Deep Operator Network [28,65]—capture nonlinear degradation effects and accelerate computational fluid dynamics and structural simulations. In practice, PINNs can infer unmeasured combustion parameters (e.g., air–fuel ratio, combustion efficiency), while ANNs optimized with metaheuristics such as Particle Swarm Optimization (PSO) enable rapid multi-output predictions of turbine inlet temperature, specific fuel consumption, and power output for real-time decision support [5]. Vibration signals provide an additional information channel. Time-domain

data are transformed into frequency-domain features such as Fast Fourier Transform (FFT) spectra, order tracking, cepstrum, and envelope analysis. These features, fused with thermo-aero residuals, allow hybrid models to capture both gas-path and rotor-dynamic degradation signatures, strengthening anomaly detection and predictive maintenance capabilities. For CFD surrogates, validation is performed not only against numerical training datasets but also through analytical or benchmark Navier–Stokes test cases, ensuring that surrogate predictions remain physically consistent and reliable [70].

4.3.3. Robustness and Uncertainty Layer

To strengthen reliability under sparse or rare-event conditions, this layer addresses data augmentation and uncertainty quantification. Generative models such as GANs, VAEs, and RNNs [8] expand training datasets, while physics-informed data generation and model discovery methods (e.g., PySINDy [24], sparse PDE learning [30], and SINDy [69]) extract interpretable governing equations for unseen scenarios. Uncertainty-aware approaches further improve robustness. Bayesian PINNs [71], Bayesian deep learning methods such as Monte Carlo dropout [77], ensemble learning [78], and recent surveys of uncertainty quantification in deep learning [79] provide calibrated confidence bounds for predictions. These probabilistic methods enhance operator trust, guide adaptive data acquisition, and supply actionable confidence levels for decision-making in digital twin deployment. Vibration feature extraction has also been widely studied for reliability improvement. Frequency-domain transformations such as FFT, order tracking, cepstrum, and envelope analysis have been applied to turbomachinery diagnostics and prognostics [80]. To overcome data scarcity, open repositories provide valuable benchmarks. The Case Western Reserve University (CWRU) Bearing Dataset [81] and NASA's Prognostics Data Repository (PDR) [82] are frequently used for validation and transfer learning, while synthetic fault signatures from physics-based models further augment sparse datasets to ensure physical consistency and generalizability.

4.3.4. Optimization and Intelligence Layer

At the highest level, embedded optimization and adaptive control modules enable predictive maintenance, virtual sensing, and fleet-wide decision-making. Low-latency optimization methods such as PSO, GA, and quadratic programming integrate with the digital twin workflow to recommend constraint aware set point adjustments and maintenance actions. As demonstrated in [5], these optimization loops can recover lost performance during alarm conditions while meeting industrial requirements for safety and responsiveness.

These integrated layers bridge design phase simulations with real-time operational intelligence, creating a framework that is interpretable, scalable, and physics consistent. By accurately capturing nonlinear behaviors and degradation mechanisms, the proposed architecture is particularly well suited to gas turbines, where high-fidelity physics and robust adaptability are essential for dependable operation. Overcoming persistent barriers—especially data scarcity, integration complexity, and computational cost—will require coordinated efforts across research communities, industrial partners, and standardization bodies.

The proposed framework aligns with emerging industrial implementations of intelligent digital twins, demonstrating both scalability and practical relevance. The EPRI turbine digital twin [56], designed for operator-focused integrated diagnostics, exemplifies how hybrid AI can be embedded into utility-scale platforms to enhance decision support. Similarly, the GE Vernova case [57] and the Siemens ATOM platform [58] illustrate fleet-wide deployment of intelligent digital twins, where cloud-enabled architectures and embedded AI modules deliver predictive maintenance, lifecycle optimization, and coordi-

nated decision-making. These case studies validate the layered hybrid AI approach and highlight the pathway toward widespread adoption of intelligent digital twins in the gas turbine industry.

These case studies validate the layered hybrid AI approach and highlight the pathway toward widespread adoption of intelligent digital twins in the gas turbine industry. To further guide research and deployment, future trends are prioritized into near-term directions such as standardized benchmarking protocols; medium-term developments including physics-constrained neural networks and CFD surrogates; and long-term research in generative approaches and operator-learning methods. This prioritization reflects both feasibility and urgency, balancing immediate industrial readiness with emerging opportunities.

5. Limitations

While this review provides a comprehensive overview of hybrid AI methods for intelligent gas turbine digital twins, it is subject to several limitations that define its scope and potential biases.

First, much of the reported progress relies on secondary sources, with synthetic datasets or simulation-based degradation models that may not fully represent actual operating conditions [41,62]. Second, the lack of widely adopted open benchmarks constrains direct comparison across hybrid AI methods and limits reproducibility [17,71]. Third, while generative and model discovery approaches such as Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and Sparse Identification of Nonlinear Dynamics (SINDy) show promise for data augmentation and rare-event prediction, their maturity in gas turbine applications remains low, with demonstrations largely confined to synthetic or small-scale test cases [8,24,30,69]. Fourth, the inherent complexity of gas turbine systems makes it challenging to develop models that simultaneously achieve fidelity, interpretability, and computational efficiency [1,2,12,62]. Finally, a broader challenge persists: realizing an intelligent digital twin that is interpretable, scalable, cyber-secure, and seamlessly integrated into industrial control and monitoring environments will require advances beyond the current state of the art [56–58].

These limitations highlight both the boundaries of the present survey and the directions where future research should focus, including benchmarking, data availability, physics-constrained methods, and integration with industrial workflows.

6. Conclusions

This review proposes a classification and a novel maturity framework for hybrid AI approaches in intelligent gas turbine digital twins, highlighting the trade-offs among ANN-augmented thermodynamic models, physics-integrated operational architectures, physics-constrained neural networks with CFD surrogates, and generative/model discovery approaches. The comparative analysis indicates that hybrid AI offers pathways to improved diagnostic accuracy, computational efficiency, and robustness compared with purely data-driven or purely physics-based methods, although adoption remains constrained by data availability, integration complexity, and validation gaps.

Based on this review, several priorities can be identified;

- Establishing open and reproducible benchmark datasets for gas turbine applications;
- Advancing uncertainty quantification methods such as Bayesian deep learning with physics-informed constraints;
- Developing validation workflows that integrate thermodynamic and CFD simulations;
- Building deployment frameworks that align with SCADA and condition monitoring infrastructures;
- Adopting cybersecurity and compliance standards for digital twin integration;

- Designing operator-facing dashboards that combine predictions with actionable confidence levels.

The classification and novel maturity framework outlined here provide actionable guidance and chart a roadmap for advancing scalable, interpretable, and operationally robust intelligent digital twins for gas turbines.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial Intelligence
ANN	Artificial Neural Network
ATOM	Autonomous Turbine Operation and Maintenance
CFD	Computational Fluid Dynamics
CHP	Combined Heat and Power
CMS	Condition Monitoring System
EPRI	Electric Power Research Institute
FNO	Fourier Neural Operator
GA	Genetic Algorithm
GAN	Generative Adversarial Network
GE	General Electric
GNN	Graph Neural Network
GPA	Gas Path Analysis
LNN	Lagrangian Neural Network
ML	Machine Learning
NPSS	Numerical Propulsion System Simulation
NSFnet	Navier–Stokes Flow Network
PcNN	Physics-Constrained Neural Network
PDE	Partial Differential Equation
PINN	Physics-Informed Neural Network
PIML	Physics-Informed Machine Learning
PSO	Particle Swarm Optimization
PySINDy	Python implementation of Sparse Identification of Nonlinear Dynamics
QP	Quadratic Programming
RNN	Recurrent Neural Network
RUL	Remaining Useful Life
SCADA	Supervisory Control and Data Acquisition
SFC	Specific Fuel Consumption
SINDy	Sparse Identification of Nonlinear Dynamics
TIT	Turbine Inlet Temperature
VAE	Variational Autoencoder

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